Bangla Distill — System Efficiency Benchmark

SUITE 3: HARDWARE PROFILE, LATENCY & EDGE DEPLOYMENT FEASIBILITY

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Target GPU: NVIDIA GeForce RTX 4070 Laptop GPU

PEAK VRAM FOOTPRINT	MIN TTFT LATENCY	PEAK THROUGHPUT	LORA ADAPTER SIZE
995.1 MB	78.5 ms	13.5 tok/s	33.6 MB
< 1.5 GB Total GPU Allocation	Real-Time First Token Delivery	~4x Higher than 9B Teachers	94.3% Storage Reduction

Edge & Mobile Feasibility Summary

Frontier teacher models (Gemma-2-9B and Qwen-3.5-9B) require dedicated data-center GPUs (A100 / RTX 3090/4090 with 20+ GB VRAM). In contrast, the distilled student model executes comfortably within **1.4 GB VRAM**, allowing it to run smoothly on edge computing boards (Jetson Orin Nano, Raspberry Pi 5 with llama.cpp, Apple Silicon unified memory) and free-tier cloud GPUs (Google Colab / Kaggle T4).

Teacher vs. Distilled Student: Footprint & Hardware Comparison

MODEL ARCHITECTURE	TOTAL PARAMS	WEIGHT STORAGE	MIN FP16 VRAM	AVG THROUGHPUT	EDGE / LAPTOP FEASIBILITY
Gemma-2-9B-IT (Teacher 1)	9.2 Billion	18.4 GB	~19.5 GB	~22 tok/s	Requires Cloud / Server GPU
Qwen-3.5-9B (Teacher 2)	9.0 Billion	18.0 GB	~19.0 GB	~23 tok/s	Requires Cloud / Server GPU
Bangla Distill Student (0.5B + LoRA)	**0.49 Billion**	**1.05 GB base + 35 MB**	**~1.4 GB VRAM**	**13.5 tok/s**	✅ **Runs on Consumer GPU & Edge**

Time To First Token (TTFT) by Prompt Size

PROMPT LENGTH	TTFT LATENCY	INTERACTIVE RATING
16 tokens	**90.42 ms**	Interactive
64 tokens	**100.73 ms**	Interactive
128 tokens	**83.71 ms**	Interactive
256 tokens	**78.48 ms**	Interactive

Throughput & VRAM Scaling by Output Length

OUTPUT LENGTH	THROUGHPUT	PEAK VRAM
32 tokens	**12.9 tok/s**	992.4 MB
64 tokens	**12.9 tok/s**	992.8 MB
128 tokens	**13.5 tok/s**	993.6 MB
256 tokens	**12.3 tok/s**	995.1 MB